Boost converter (12 V)

L1 MWSA0402S-100MT

U1 TLV61046A (SW, VIN, GND, VO, FB, EN)

C2 4.7u, C3 10u

U1 TLV61046A boost from +5V (USB VBUS). FB and EN tied to VIN: internal fixed 12 V output, always enabled.
L1 10 uH, C2 4.7u in, C3 10u out.
12 V out feeds the Q2 keying switch.

Keying

TX (MCU PA4) drives Q1 2N7002; R1 22k pull-down holds it off at reset.
Q1 pulls Q2 MMBT4403 base via R3 22k; high-side switch for the 12 V PA supply. R2 22k B-E pull-up, C4 10n C-B.
R6 10R + C7 1u filter.
R8 1R 1% PA current sense: measure across J2 PAI+ / J3 PAI-.

Q2 MMBT4403, R2 22k, R3 22k, C4 10n, R6 10, R8 1 1%, C7 1u, J2 PAI+, J3 PAI-, Q1 2N7002, R1 22k, TX

Power amplifier

CLK0 (SI5351C) -> U3 SN74LVC1G17 buffer -> C5 100n -> Q3/Q8 BS170 in parallel (DNP, hand-solder), R16/R17 10R gate stoppers.
Drain fed from J3 via L2. R5 3.3k + C8 100n: drain to buffer-out RC. C11 100n DC block to LPF_IN.

U3 SN74LVC1G17, C1 100n, C5 100n, R5 3.3k, C8 100n, C11 100n, L2 FB2402-61 5T, R16 10, R17 10, R7 56, Q3 BS170, Q8 BS170, CLK0

PA bias

U5 MCP4725 12-bit DAC on SCL_5V / SDA_5V (A0 = GND).
VOUT -> R7 56R -> PA gates.
C9 1u on VOUT, C10 100n VDD.
J1: bias voltage test point.

MCP4725A1T-E/CH U5 (VOUT A0, VSS SCL, VDD SDA), J1 Bias V, C9 1u, C10 100n, SCL_5V, SDA_5V

Si5351C synthesizer

R4 10, C6 100n, U2 25.000 MHz TCXO, C12 100n, C14 100n, C50 100n, C51 100n, C52 100n, C53 100n

J16 SMA CLKIN, CLKIN_SMA, C54 100n, D21 PESD5V0F1BL, R43 49.9, R44 10k, R45 10k, R46 10k, SI_INTR

U4 SI5351C-B-GMR (XA, XB, CLKIN, SDA, SCL, INTR, OEB, VDD, VDDOA, VDDOB, VDDOC, VDDOD, GND, CLK0, CLK1, CLK2, CLK3, CLK4, CLK5, CLK6, CLK7)

J6 SMA, R47 33, CAL_CLK, CLK0

XA: U2 25 MHz TCXO, R4 10R feed, R13 OR OE (DNP).
C12 100n to XA; XB open.
J16 CLKIN (10-100 MHz): R43 49.9R termination, D21 PESD5V0F1BL ESD, C54 100n AC coupling.
R44/R45 10k/10k bias at VDD/2.
OEB tied to GND; outputs enabled over I2C.

RF output: low-pass filter bank (sheet 3)

Filter Bank (BAND_A, BAND_B, LPF_IN, LPF_OUT)

File: filter_bank.kicad_sch

BAND_A/BAND_B select one of four LPFs:
00 80/60 m, 01 40/30 m, 10 20/17 m, 11 15/12/10 m.
J11 BNC output (DNP, hand-soldered).

J11 BNC

MCU + USB

File: mcu_usb.kicad_sch

GPS

File: gps.kicad_sch

I/O + UI

File: io_ui.kicad_sch

J9 GND

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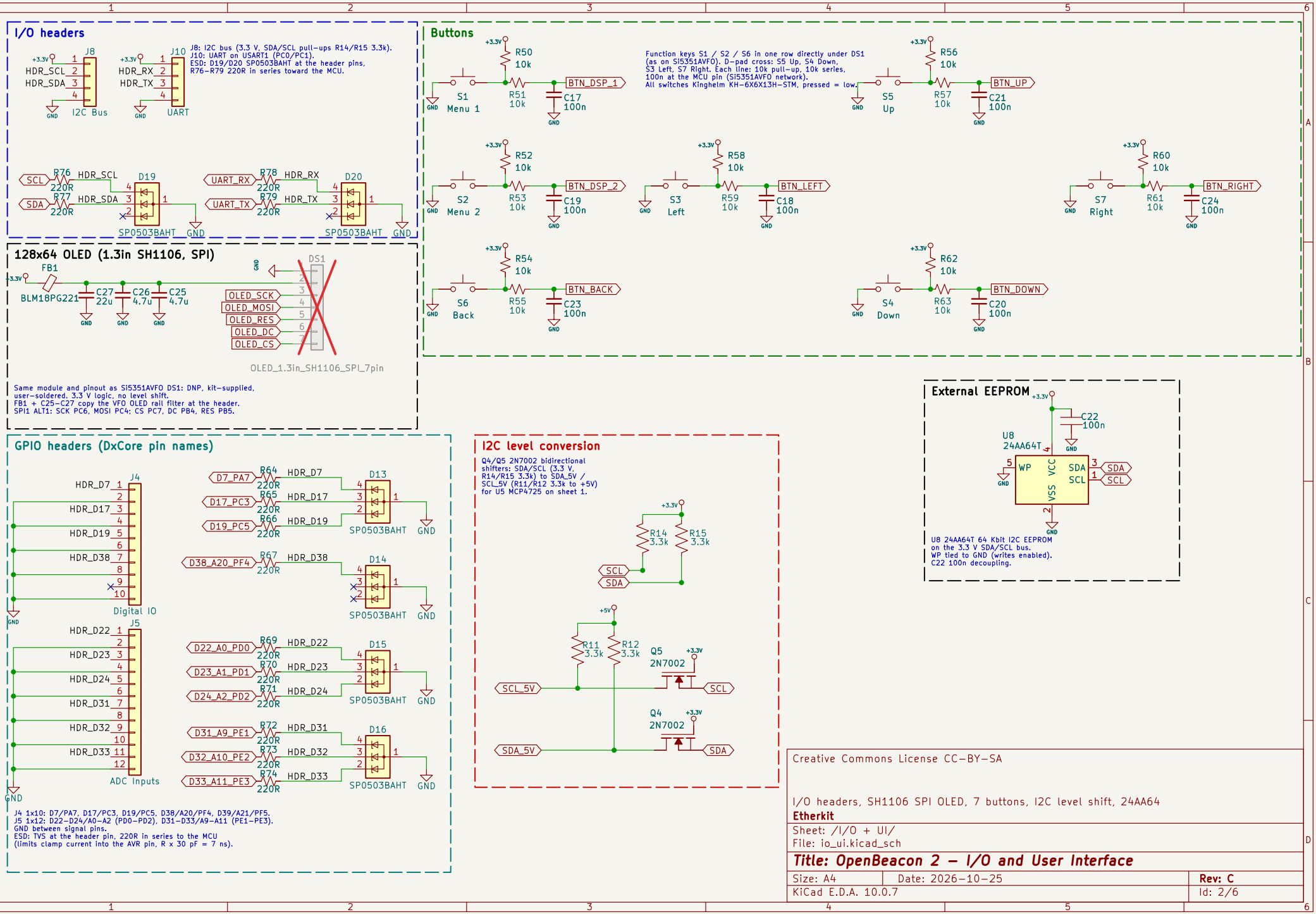
RF chain: boost, keying, PA, bias DAC, Si5351C synth, filter bank, BNC out

Etherkit

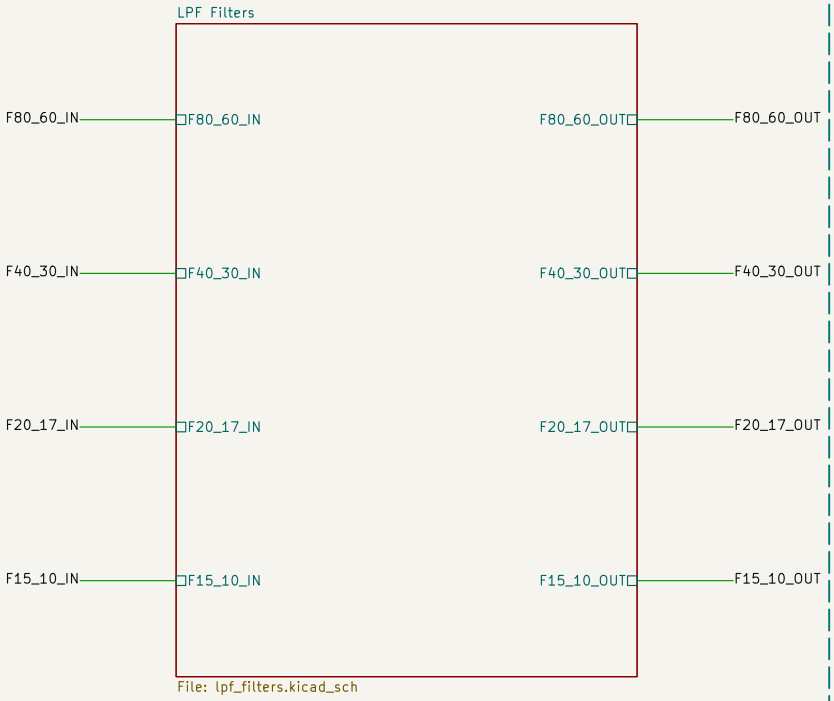
Sheet: /
File: OpenBeacon2.kicad_sch

Title: OpenBeacon 2

Size: A4	Date: 2026-10-25	Rev: C
KiCad E.D.A. 10.0.7		Id: 1/6

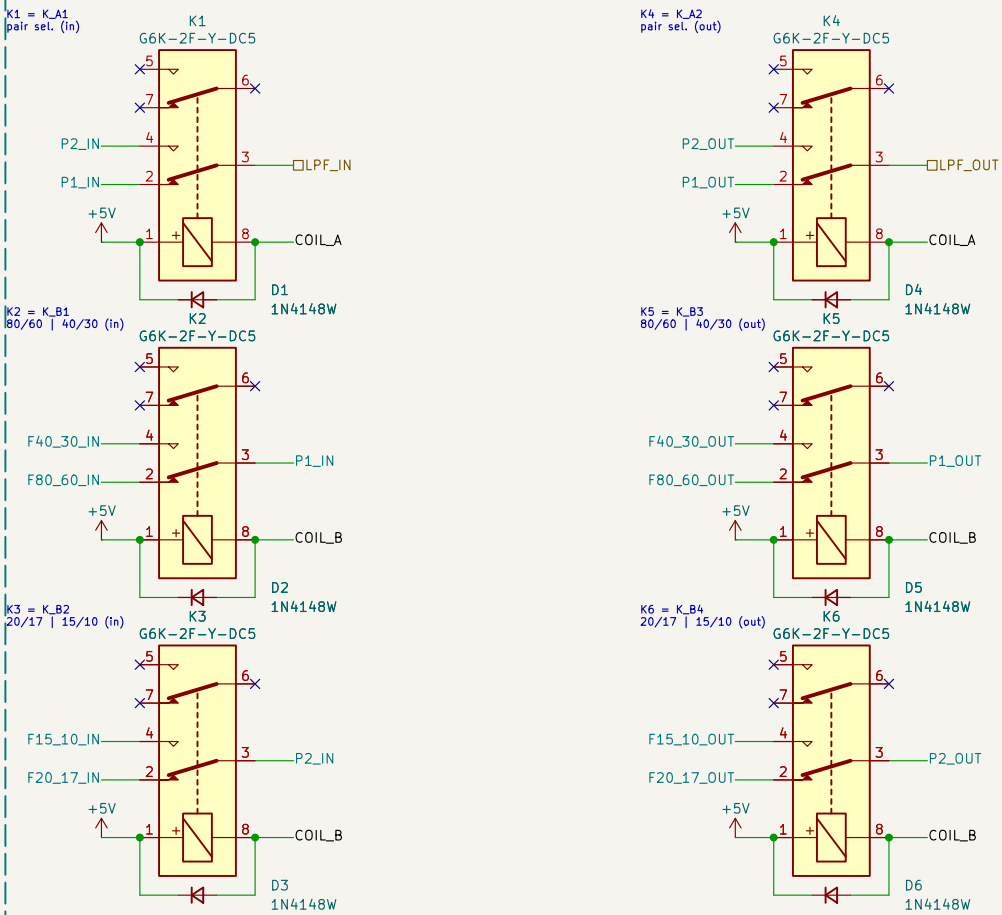

I/O headers
J8: I2C bus (3.3 V, SDA/SCL pull-ups R14/R15 3.3k).
J10: UART on USART1 (PC0/PC1).
ESD: D19/D20 SP0503BAHT at the header pins,
R76–R79 220R in series toward the MCU.
J8
HDR_SCL
HDR_SDA
I2C Bus
J10
HDR_RX
HDR_TX
UART
SCL
SDA
R76 220R
R77 220R
HDR_SCL
HDR_SDA
D19
SP0503BAHT
UART_RX
UART_TX
R78 220R
R79 220R
HDR_RX
HDR_TX
D20
SP0503BAHT
GND
128x64 OLED (1.3in SH1106, SPI)
FB1
BLM18PG221
C27 22u
C26 4.7u
C25 4.7u
DS1
OLED_SCK
OLED_MOSI
OLED_RES
OLED_DC
OLED_CS
OLED_1.3in_SH1106_SPI_7pin
Same module and pinout as SI5351AVFO DS1: DNP, kit-supplied,
user-soldered, 3.3 V logic, no level shift.
FB1 + C25–C27 copy the VFO OLED rail filter at the header.
SPI1 ALT1: SCK PC6, MOSI PC4; CS PC7, DC PB4, RES PB5.
Buttons
Function keys S1 / S2 / S6 in one row directly under DS1
(as on SI5351AVFO). D-pad cross: S5 Up, S4 Down,
S3 Left, S7 Right. Each line: 10k pull-up, 10k series,
100n at the MCU pin (SI5351AVFO network).
All switches Kinghelm KH-6X6X13H-STM, pressed = low.
S1 Menu 1
S2 Menu 2
S3 Left
S4 Down
S5 Up
S6 Back
S7 Right
R50 10k
R51 10k
C17 100n
BTN_DSP_1
R52 10k
R53 10k
C19 100n
BTN_DSP_2
R54 10k
R55 10k
C23 100n
BTN_BACK
R56 10k
R57 10k
C21 100n
BTN_UP
R58 10k
R59 10k
C18 100n
BTN_LEFT
R62 10k
R63 10k
C20 100n
BTN_DOWN
R60 10k
R61 10k
C24 100n
BTN_RIGHT
External EEPROM
U8
24AA64T
WP
VCC
VSS
SDA
SCL
C22 100n
U8 24AA64T 64 Kbit I2C EEPROM
on the 3.3 V SDA/SCL bus.
WP tied to GND (writes enabled).
C22 100n decoupling.
GPIO headers (DxCore pin names)
J4
HDR_D7
HDR_D17
HDR_D19
HDR_D38
Digital IO
J5
HDR_D22
HDR_D23
HDR_D24
HDR_D31
HDR_D32
HDR_D33
ADC Inputs
D7_PA7
D17_PC3
D19_PC5
D38_A20_PF4
D22_A0_PD0
D23_A1_PD1
D24_A2_PD2
D31_A9_PE1
D32_A10_PE2
D33_A11_PE3
R64 220R
R65 220R
R66 220R
R67 220R
R69 220R
R70 220R
R71 220R
R72 220R
R73 220R
R74 220R
D13
D14
D15
D16
SP0503BAHT
J4 1x10: D7/PA7, D17/PC3, D19/PC5, D38/A20/PF4, D39/A21/PF5.
J5 1x12: D22–D24/A0–A2 (PD0–PD2), D31–D33/A9–A11 (PE1–PE3).
GND between signal pins.
ESD: TVS at the header pin, 220R in series to the MCU
(limits clamp current into the AVR pin, R x 30 pF = 7 ns).
I2C level conversion
Q4/Q5 2N7002 bidirectional
shifters: SDA/SCL (3.3 V
R14/R15 3.3k) to SDA_5V /
SCL_5V (R11/R12 3.3k to +5V)
for U5 MCP4725 on sheet 1.
R14 3.3k
R15 3.3k
R11 3.3k
R12 3.3k
Q5 2N7002
Q4 2N7002
SCL_5V
SDA_5V
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I/O headers, SH1106 SPI OLED, 7 buttons, I2C level shift, 24AA64
Etherkit
Sheet: /I/O + UI/
File: io_ui.kicad_sch
Title: OpenBeacon 2 – I/O and User Interface
Size: A4
Date: 2026-10-25
Rev: C
KiCad E.D.A. 10.0.7
Id: 2/6

F2–F5 low-pass filters (T30 toroids)

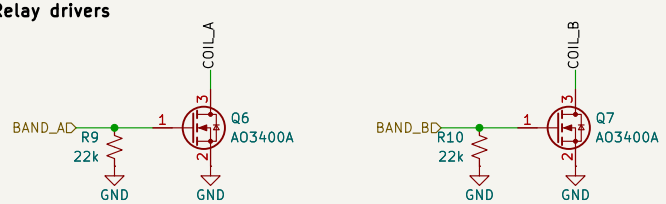


Four 7th-order Cauer LPFs, hand-wound T30 toroids.
Schematic + winding table on sheet LPF Filters.
Designed for 1 W output max.

Relay tree (2 levels, input + output mirrored; all off = 80/60 m)



Relay drivers



BAND_A -> K1/K4, BAND_B -> K2/K3/K5/K6.
Q6 (BAND_A): 2 coils, 42 mA nom (5 V / 237 R),
<= 49 mA at 5.25 V, -10 % coil R.
Q7 (BAND_B): 4 coils, 84 mA nom, <= 98 mA.
AO3400A: 5.7 A cont., 48 mR @ Vgs 2.5 V -> <5 mV drop.
One 1N4148W flyback per coil, at each relay.

Band code + notes

Band code (BAND_A, BAND_B):
0 0 80/60 m (all relays off, NC path, default)
0 1 40/30 m (K2, K3, K5, K6 on)
1 0 20/17 m (K1, K4 on)
1 1 15/12/10 m (all six on)
Unselected filters float (open at both ends).
+5V relay load: 0 / 84 / 42 / 127 mA nominal (<=147 mA).
Relay pins: COM right, NC lower left, NO upper left; pole 2 unused.
Input and output paths never share a relay body.
Layout: keep K1–K2–K3 and K4–K5–K6 traces short,
50 ohm CPWG over solid GND, filters next to their relays.

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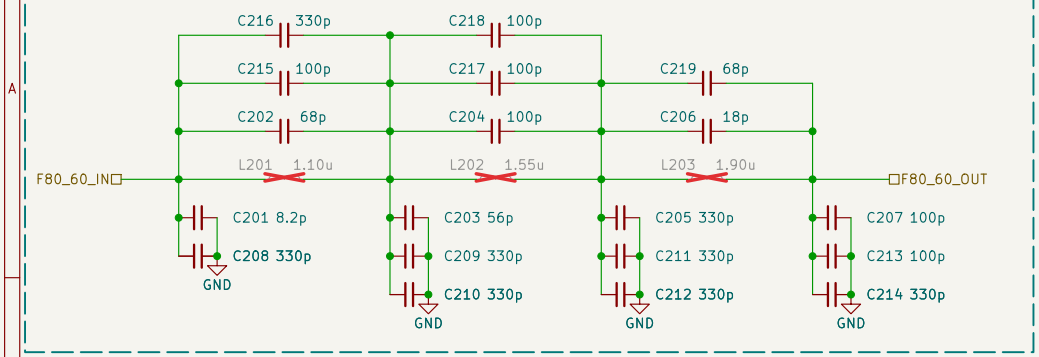
4 7th-order Cauer LPFs, 6-relay 2-level tree (BAND_A/BAND_B)

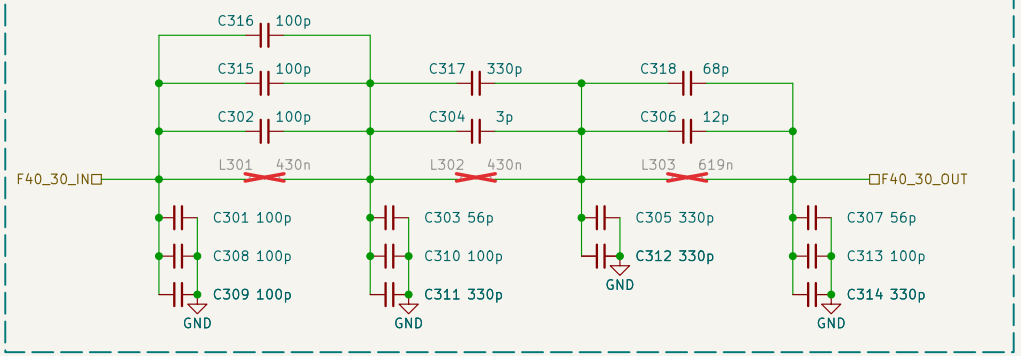
Etherkit

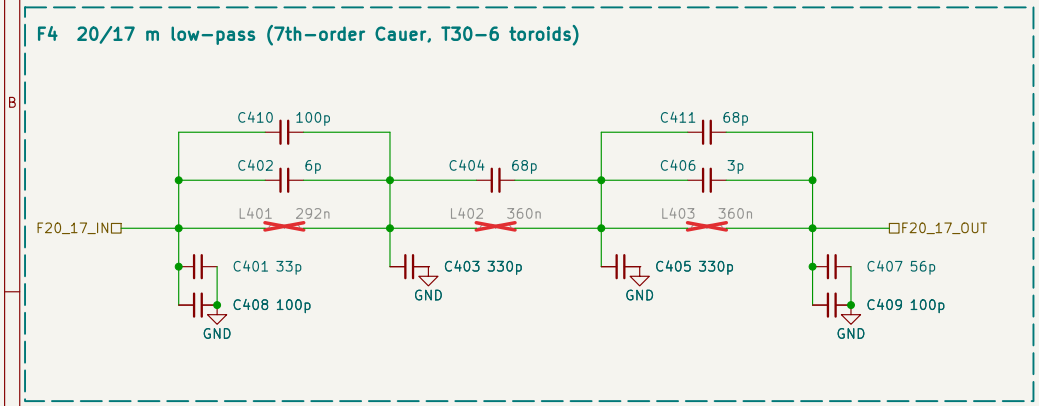
Sheet: /Filter Bank/
File: filter_bank.kicad_sch

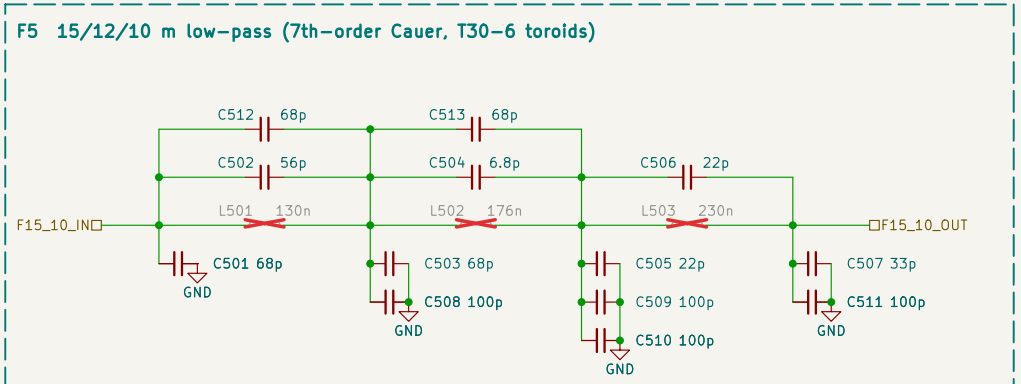
Title: OpenBeacon 2 – Low-Pass Filter Bank

Size: A4	Date: 2026-10-25	Rev: C
KiCad E.D.A. 10.0.7		Id: 3/6









Toroid winding table (hand-wound, not placed by JLC)

Ref	Filter / bands	Inductance	Core (Kits and Parts)	Turns	Wire
L201	F2 80/60 m	1.10 uH	T30-2 (red)	16	#28 AWG enam.
L202	F2 80/60 m	1.55 uH	T30-2 (red)	19	#28 AWG enam.
L203	F2 80/60 m	1.90 uH	T30-2 (red)	21	#30 AWG enam.
L301	F3 40/30 m	430 nH	T30-2 (red)	10	#24 AWG enam.
L302	F3 40/30 m	430 nH	T30-2 (red)	10	#24 AWG enam.
L303	F3 40/30 m	619 nH	T30-2 (red)	12	#26 AWG enam.
L401	F4 20/17 m	292 nH	T30-6 (yellow)	9	#24 AWG enam.
L402	F4 20/17 m	360 nH	T30-6 (yellow)	10	#24 AWG enam.
L403	F4 20/17 m	360 nH	T30-6 (yellow)	10	#24 AWG enam.
L501	F5 15/12/10 m	130 nH	T30-6 (yellow)	6	#24 AWG enam.
L502	F5 15/12/10 m	176 nH	T30-6 (yellow)	7	#24 AWG enam.
L503	F5 15/12/10 m	230 nH	T30-6 (yellow)	8	#24 AWG enam.

Cores: Micrometals T30-2 (red, F2/F3) and T30-6 (yellow, F4/F5) from Kits and Parts (kitsandparts.com).
Turns = passes through the core centre. AL 4.3 nH/t^2 (-2), 3.6 nH/t^2 (-6), +/-5%%.
F5 has few turns: tune each toroid by spreading or compressing turns. Mount toroids vertically.
Rated for 1 W output max. All filter caps 0603 C0G/NP0 50 V; parallel parts sum to the design value.

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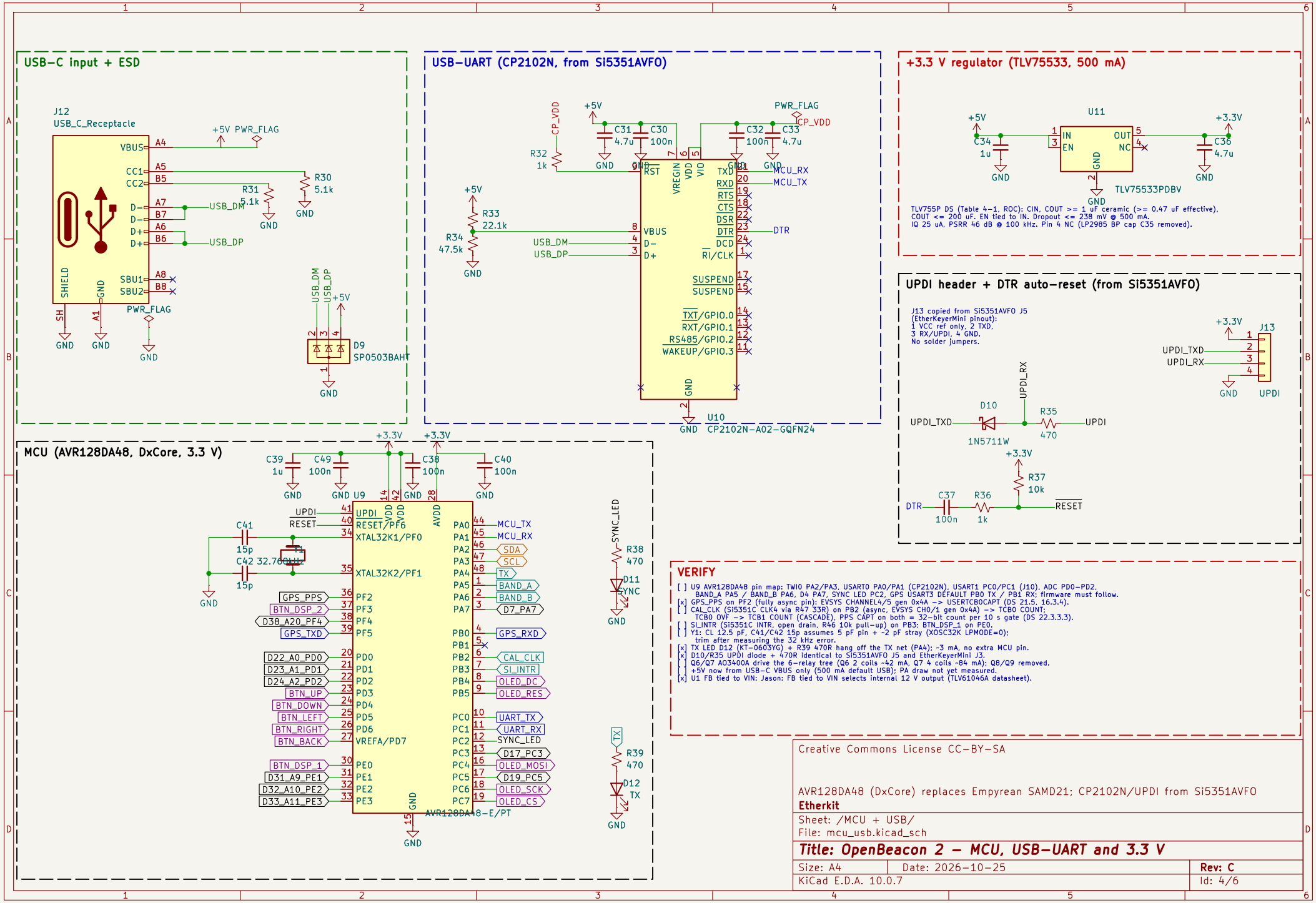
F2-F5 7th-order Cauer LPFs on hand-wound T30 toroids, 1 W max

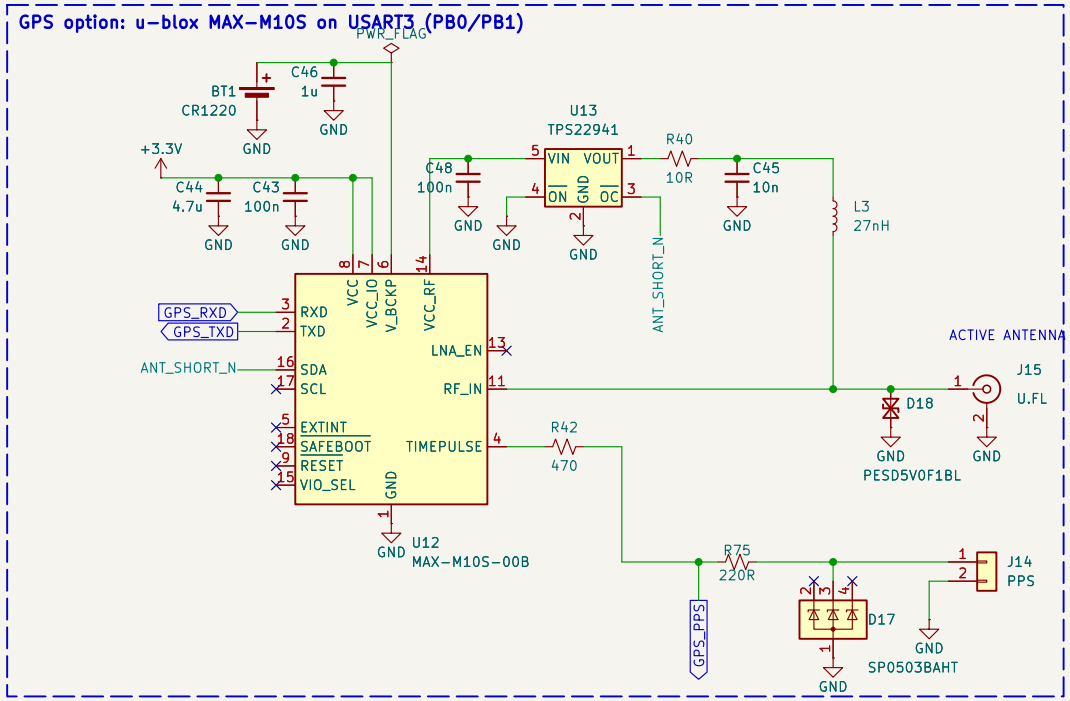
Etherkit

Sheet: /Filter Bank/LPF Filters/
File: lpf_filters.kicad_sch

Title: OpenBeacon 2 - LPFs F2-F5 (T30 toroids)

Size: A4	Date: 2026-10-25	**Rev: C**
KiCad E.D.A. 10.0.7		Id: 6/6


USB-C input + ESD
J12
USB_C_Receptacle
+5V
PWR_FLAG
VBUS
CC1
CC2
D-
D-
D+
D+
SHIELD
GND
SBU1
SBU2
A4
A5
B5
A7
B7
A6
B6
A8
B8
R30
5.1k
R31
5.1k
USB_DM
USB_DP
SH
A1
D9
SP0503BAHT
USB-UART (CP2102N, from Si5351AVFO)
CP_VDD
C31
4.7u
C30
100n
C32
100n
C33
4.7u
R32
1k
RST
VREGIN
VDD
VIO
TXD
RXD
RTS
CTS
DSR
DTR
DCD
RI/CLK
SUSPEND
TXT/GPIO.0
RXT/GPIO.1
RS485/GPIO.2
WAKEUP/GPIO.3
MCU_RX
MCU_TX
DTR
R33
22.1k
R34
47.5k
VBUS
D-
D+
U10
CP2102N-A02-GQFN24
+3.3 V regulator (TLV75533, 500 mA)
U11
C34
1u
C36
4.7u
IN
EN
OUT
NC
TLV75533PDBV
TLV755P DS (Table 4-1, ROC): CIN, COUT >= 1 uF ceramic (>= 0.47 uF effective),
COUT <= 200 uF. EN tied to IN. Dropout <= 238 mV @ 500 mA.
IQ 25 uA, PSRR 46 dB @ 100 kHz. Pin 4 NC (LP2985 BP cap C35 removed).
UPDI header + DTR auto-reset (from Si5351AVFO)
J13 copied from Si5351AVFO J5
(EtherKeyerMini pinout):
1 VCC ref only, 2 TXD,
3 RX/UPDI, 4 GND.
No solder jumpers.
J13
UPDI_TXD
UPDI_RX
UPDI
D10
1N5711W
UPDI_RX
R35
470
R37
10k
C37
100n
R36
1k
DTR
RESET
MCU (AVR128DA48, DxCore, 3.3 V)
C39
1u
C49
100n
C38
100n
C40
100n
U9
UPDI
RESET/PF6
XTAL32K1/PF0
XTAL32K2/PF1
C41
15p
C42
15p
Y1
32.768kHz
PF2
PF3
PF4
PF5
PD0
PD1
PD2
PD3
PD4
PD5
PD6
VREFA/PD7
PE0
PE1
PE2
PE3
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PB0
PB1
PB2
PB3
PB4
PB5
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
AVR128DA48-E/PT
GPS_PPS
BTN_DSP_2
D38_A20_PF4
GPS_TXD
D22_A0_PD0
D23_A1_PD1
D24_A2_PD2
BTN_UP
BTN_DOWN
BTN_LEFT
BTN_RIGHT
BTN_BACK
BTN_DSP_1
D31_A9_PE1
D32_A10_PE2
D33_A11_PE3
MCU_TX
MCU_RX
SDA
SCL
TX
BAND_A
BAND_B
D7_PA7
GPS_RXD
CAL_CLK
SI_INTR
OLED_DC
OLED_RES
UART_TX
UART_RX
SYNC_LED
D17_PC3
OLED_MOSI
D19_PC5
OLED_SCK
OLED_CS
R38
470
D11
SYNC
R39
470
D12
TX
VERIFY
[ ] U9 AVR128DA48 pin map: TWI0 PA2/PA3, USART0 PA0/PA1 (CP2102N), USART1 PC0/PC1 (J10), ADC PD0-PD2,
BAND_A PA5 / BAND_B PA6, D4 PA7, SYNC LED PC2, GPS USART3 DEFAULT PB0 TX / PB1 RX; firmware must follow.
[x] GPS_PPS on PF2 (fully async pin): EVSYS CHANNEL4/5 gen 0x4A -> USERTCBOCAPT (DS 21.5, 16.3.4).
[ ] CAL_CLK (SI5351C CLK4 via R47 33R) on PB2 (async, EVSYS CH0/1 gen 0x4A) -> TCB0 COUNT;
TCB0 OVF -> TCB1 COUNT (CASCADE), PPS CAPT on both = 32-bit count per 10 s gate (DS 22.3.3.3).
[ ] SI_INTR (SI5351C INTR, open drain, R46 10k pull-up) on PB3; BTN_DSP_1 on PE0.
[ ] Y1: CL 12.5 pF, C41/C42 15p assumes 5 pF pin + ~2 pF stray (XOSC32K LPMODE=0);
trim after measuring the 32 kHz error.
[x] TX LED D12 (KT-0603YG) + R39 470R hang off the TX net (PA4); ~3 mA, no extra MCU pin.
[x] D10/R35 UPDI diode + 470R identical to Si5351AVFO J5 and EtherKeyerMini J3.
[ ] Q6/Q7 AO3400A drive the 6-relay tree (Q6 2 coils ~42 mA, Q7 4 coils ~84 mA); Q8/Q9 removed.
[ ] +5V now from USB-C VBUS only (500 mA default USB); PA draw not yet measured.
[x] U1 FB tied to VIN: Jason: FB tied to VIN selects internal 12 V output (TLV61046A datasheet).
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AVR128DA48 (DxCore) replaces Empyrean SAMD21; CP2102N/UPDI from Si5351AVFO
Etherkit
Sheet: /MCU + USB/
File: mcu_usb.kicad_sch
Title: OpenBeacon 2 - MCU, USB-UART and 3.3 V
Size: A4
Date: 2026-10-25
Rev: C
KiCad E.D.A. 10.0.7
Id: 4/6



VERIFY

- [x] Antenna short: TPS22941 limit 40–80 mA, auto-restart, avg <= 27 mA (was 320 mA).
- [x] V_BCKP: CR1220 in KH-BS1220-2-SMT (C5365932): C47/D13/R41 removed.
- [ ] BT1 footprint: draw from KH-BS1220-2-SMT DS.
- [ ] U13 TPS22941 JLC stock thin (26 pcs).
- [ ] Short peak 80 mA > VCC_RF 50 mA op. (DS T13) for <= 20 ms bursts; confirm on bench.
- [ ] V_IO ramp 25–35000 us/V (DS 4.1): measure.
- [ ] GPS_PPS MCU pin: see mcu_usb.

RECEIVER: VCC + V_IO tied to +3.3V; VIO_SEL open = 3.3 V I/O (IM 4.1.2).
C43/C44 beside pins 7/8, short GND vias; no series R > 0.2 ohm in VCC (IM 4.1.1).
RESET_N, EXTINT, SCL, SAFEBOOT_N, LNA_EN left open (DS Table 10).
SDA = ANT_SHORT_N (IM 3.3.1.2): CFG-HW-ANT_CFG_SHORTDET=1, ANT_SUP_SHORT_PIN=SDA PIO.
PPS: TIMEPULSE = SAFEBOOT_N via 1k inside (DS fn 15); nothing may hold it low at power-up, so no pull-down. R42 limits a shorted J14 to ~7 mA (I_PIO 10 mA).

ANTENNA: RF_IN is DC-blocked inside the module (IM Table 1), no external DC block.
Bias-T VCC_RF -> U13 TPS22941 -> R40 10R -> C45 10n -> L3 27 nH onto RF line (IM 3.3.1.2, 4.3.4).
U13: I_LIM 40–80 mA, 5–20 ms blanking, 40–160 ms restart (TPS2294x DS 7.5): short avg <= 27 mA, typ 7 mA (VCC_RF abs max 250 mA, DS Table 12). 25 mA antenna drop ~0.27 V -> ~2.9 V at J15.
Layout: 50 ohm CPWG pin 11 -> J15, L3 pad on the line, >= 5 mm from digital.
BACKUP: BT1 CR1220 (3 V, 35 mAh, user-fitted, NOT rechargeable, no charger) on V_BCKP;
C46 1u at pin 6 (IM 4.1.3). 35 mAh / 28 uA (DS Table 17) = ~1250 h (~52 days) powered off.

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MAX-M10S on USART3 (PB0/PB1), PPS to J14, active-antenna bias-T, CR1220

Etherkit

Sheet: /GPS/
File: gps.kicad_sch

Title: OpenBeacon 2 – GPS (u-blox MAX-M10S)

Size: A4	Date: 2026-10-25	**Rev: C**
KiCad E.D.A. 10.0.7		Id: 5/6